EXTREMAL GRAPH THEORY

Erdős Problem 915

From Proof to Search

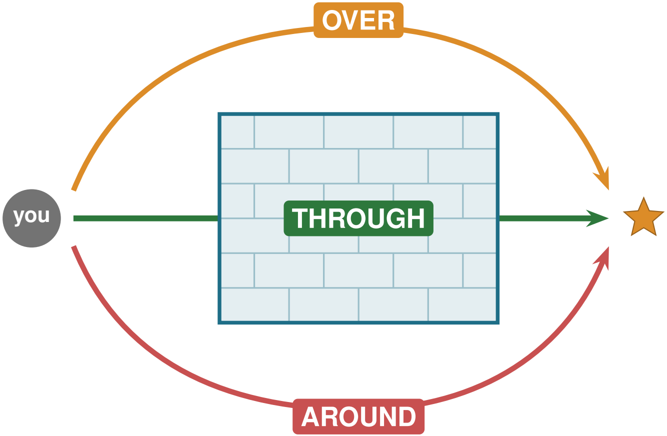
Louis Vandenbruwaene

Supervisor: Prof. Stijn Cambie Promotor: Prof. Jan Goedgebeur

KU Leuven, Master of Mathematics

KU LEUVEN

A hard problem is a wall

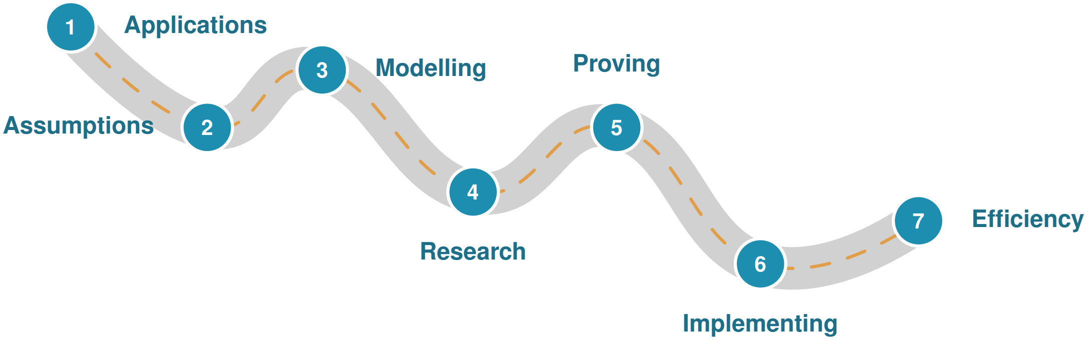


Three ways past the wall

THROUGH	OVER	AROUND
Proof	**Calculation**	**Approximation**
one argument, every n at once	a finite check, with a certificate	search for examples, a lower bound

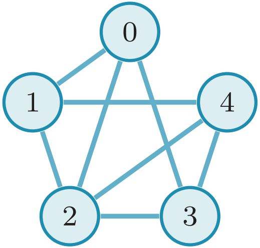
This thesis uses all three.

The road of a research problem



The slides ahead follow this road.

Problem 915, precisely



K_5 minus 2 edges: $8 = \ell_4(5)$ edges

A graph: n vertices, E edges.

Routes. $\lambda(u, v)$ = independent u-v paths.

Rule. No pair gets m routes: $\lambda^{\max} \leq m - 1$.

Goal. Pack in as many edges as possible.

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor \quad \text{(Mader)}$$

What is it good for?

No direct application. The methods are the point.

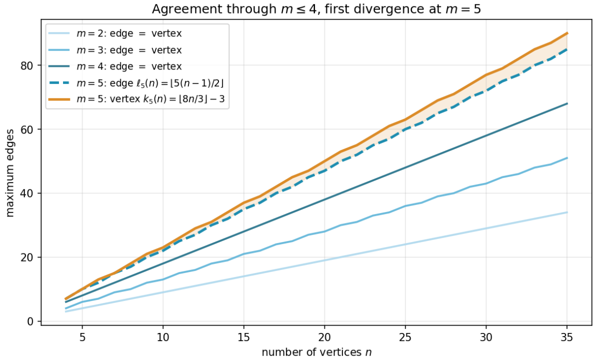
The question	The tools	Where it lives
network redundancy	flow, search, proofs	grids, pipelines, roads

One question, twelve variants

Model	simple	multigraph	hypergraph	$\times 3$	
Direction	undirected	directed		$\times 2$	$\boxed{= 12}$
Separation	edge-disjoint	vertex-disjoint		$\times 2$	

Mader's theorem is one corner. The other eleven are the journey.

Edge routes versus vertex routes



λ: edge-disjoint. κ: vertex-disjoint.

$m \le 4$ same answer (Leonard).

$m = 5$ first gap: $k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 3$.

$m \ge 6$ open since 1974.

What fell, and how hard

Proved by hand	Certified by machine	Conjecture + lower bound	Open
undirected edge	directed multigraph $n \leq 6$	directed arc $m \geq 3$	vertex-disjoint $m \geq 6$
multigraph vertex	hypergraph vertex $m \leq 3$	directed vertex $m \geq 3$	hypergraph vertex $m \geq 4$
directed $m = 2$			directed hypergraph

easier → harder

How the machine works

Which search wins

variant	best	SA	tabu
simple undirected $n=7$	9	9	$\mathbf{9}$
simple directed $n=7$	18	16	$\mathbf{18}$
directed multi $n=7$	24	20	$\mathbf{24}$

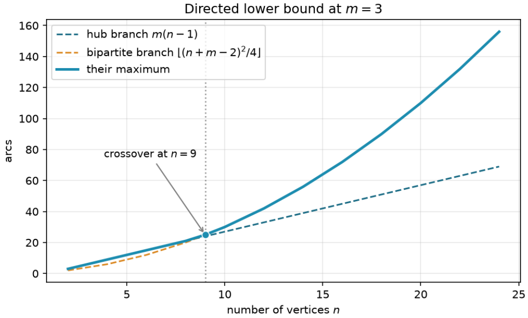
Tabu reaches the hard directed optima.

What makes it fast

C hot-path routine	speedup
dense max-flow	$2\times$
connectivity test	$2.2\times$
canonical form	$\mathbf{147}\times$

Python core, optional C, same answers.

New results: hypergraph vertex and directed arcs



The quadratic family overtakes near $n = 9$.

Two genuine advances

solved

Hypergraph vertex $(m \le 3)$

$$k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$$

reduced

Directed multigraph $(m = 3)$

all n $\longrightarrow$ two checks at $n{=}7$

The whole landscape

Separation	Model	Value	Status
Undirected			
edge	simple	$\lfloor m(n-1)/2 \rfloor$	proved (Mader)
edge	multi	$(m-1)(n-1)$	proved
edge	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$	proved (Gomory-Hu)
vertex	simple	$\lfloor m(n-1)/2 \rfloor$ $(m \leq 4)$, $\lfloor 8n/3 \rfloor - 3$ $(m{=}5)$	cited, open $m \geq 6$
vertex	multi	parallel edges do not help	proved
vertex	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$ $(m \leq 3)$	proved $m \leq 3$, open $m \geq 4$
Directed			
arc	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ $(m{=}2)$	proved, conj. $m \geq 3$
arc	multi	$(m-1)\max(2(n-1), \lfloor n^2/4 \rfloor)$	proved $n \leq 6$, cond. $m{=}3$
arc	hyper	quadratic in n, exact value open	lower bound
vertex	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ $(m{=}2)$	proved, conj. $m \geq 3$
vertex	multi	reduces to simple digraph	proved
vertex	hyper	quadratic in n, exact value open	lower bound

proved **certified** **conjecture** **open**

What is left, and why it matters

Open problems

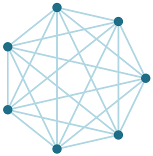
- the backward-arc lemma
- the linear coefficient, one-way
- vertex-disjoint, $m \geq 6$
- hypergraph vertex, $m \geq 4$

One pipeline, every variant

- one representation
- one exact checker
- one certifier
- one search

Thank you

Questions?

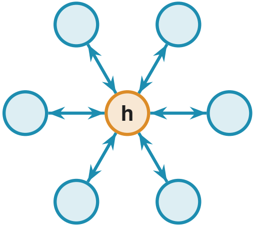


Backup slides follow.

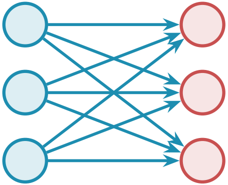
BACKUP SLIDES

Dive deeper into any single variant.

Backup: the directed arc conjecture



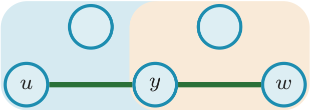
bidirected hub: $2(n-1)$



bipartite digraph: $\lfloor n^2/4 \rfloor$

At $m = 2$ the value $\max(2(n-1), \lfloor n^2/4 \rfloor)$ is proved (the two constructions tie at $n = 7$). For $m \geq 3$ the conjecture adds the augmented-bipartite family. The missing piece is the **backward-arc lemma** for the matching upper bound.

Backup: the hypergraph vertex problem



a Berge path u, e_1, y, e_2, w

Routes are **Berge paths**, alternating vertices and hyperedges.

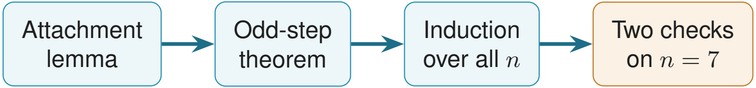
Solved ($m \le 3$, every r, all n):

$$k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

via a rank bound on incidence graphs (Tutte SPQR, valid where $\kappa \le 2$).

Open at $m \ge 4$**:** needs a 4-connectivity analogue.

Backup: collapsing an infinite family to two checks



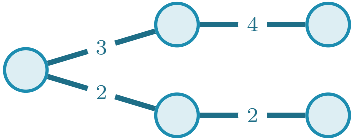
The two remaining facts (multiplicities in $\{0, 1, 2\}$):

(a) $L_3^{\mathrm{dir}}(7) = 24$;

(b) the only 24-arc extremisers are $2B_{3,4}$, $2B_{4,3}$, and the doubled path P_7.

The wall here is runtime, not structure.

Backup: how connectivity is stored



A **Gomory-Hu tree** stores all $\binom{n}{2}$ connectivities in $n-1$ numbers: $\lambda(u,v)$ is the lightest edge on the tree path, and $\lambda^{\max}$ is the heaviest tree edge.

Backup: values at a glance

Object	Extremal value
simple undirected edge, $\ell_m(n)$	$\lfloor m(n-1)/2 \rfloor$
multigraph edge, $L_m(n)$	$(m-1)(n-1)$
hypergraph edge, $\ell_m^{(r)}(n)$	$\lfloor (m-1)(n-1)/(r-1) \rfloor$
simple vertex, $k_m(n)$ ($m \leq 4$)	$\lfloor m(n-1)/2 \rfloor$
simple vertex, $k_5(n)$	$\lfloor 8n/3 \rfloor - 3$
directed arc, $m = 2$	$\max(2(n-1), \lfloor n^2/4 \rfloor)$
hypergraph vertex, $m \leq 3$	$\lfloor (m-1)(n-1)/(r-1) \rfloor$
random-graph threshold	$p^* = m/n$